

Understanding the Relationship Between Screen Time and Procrastination Among University Students: A Data Analytics Case Study

1.Introduction

Smartphones have become an essential part of university students' daily lives, supporting learning, communication, entertainment, and social interaction. While digital technologies provide numerous educational benefits, excessive screen time has increasingly become a concern due to its potential influence on academic performance, time management, and productivity. The widespread use of social media platforms, video streaming services, online gaming, and instant messaging applications has made digital distractions more common, often contributing to delayed academic tasks and reduced study efficiency.

Procrastination is a common behavioural challenge among university students and is often characterized by the intentional postponement of academic responsibilities despite awareness of potential negative consequences. Understanding how smartphone usage patterns relate to procrastination can help educational institutions, digital well-being platforms, and productivity application developers design strategies that encourage healthier technology use and improve student productivity.

This case study applies the data analytics process to investigate screen time behaviour among university students aged 20–30 years. Through exploratory data analysis, statistical techniques, and interactive visualizations, the study aims to identify usage patterns, examine relationships between screen time and procrastination, and generate data-driven recommendations that promote healthier digital habits and improved academic productivity.

2.Scenario

I am a Junior Data Analyst at FocusTech, a technology company that develops digital well-being and productivity applications designed to help university students manage their smartphone usage and improve academic performance.

Recently, the product management team observed that many users reported difficulty maintaining focus during study sessions due to prolonged smartphone use. Although the company collects user activity data through its application, it has limited understanding of how different screen time behaviours relate to procrastination among university students.

To support future product improvements, the analytics team has been assigned to investigate smartphone usage patterns and identify behaviours that may contribute to procrastination. My role is to analyse the available data, uncover meaningful insights, and present evidence-based recommendations that will help the product team design features promoting healthier digital habits and improved academic productivity.

(INFO: FocusTech is an educational technology (EdTech) company that develops AI-powered digital well-being and productivity solutions for university students. Its flagship mobile application helps students monitor screen time, manage distractions, track study habits, and build healthier digital routines through personalized insights and analytics.)

3. Business Task

Analyse smartphone screen time data from university students aged 20–30 years to identify behavioural patterns associated with procrastination and provide data-driven recommendations that support digital well-being and academic productivity.

4. Stakeholders

Primary Stakeholders	Secondary Stakeholders
Product Manager	University Students
Digital Well-being Team	Mobile Application Developers
Data Analytics Team	Student Success and Academic Support Teams
UX/UI Design Team	
Executive Management	

5. Business Objective

The objectives of this case study are to:

- Analyze daily smartphone screen time among university students.
- Identify the application categories contributing most to screen time.
- Examine the relationship between screen time and procrastination.
- Explore how screen time varies by academic year, study habits, and sleep duration (if the data supports these variables).
- Identify user behavior patterns that indicate excessive smartphone usage.
- Provide evidence-based recommendations for improving digital productivity and well-being.

6.Key Business Questions

Screen Time

- What is the average daily screen time of university students?
- Which application categories consume the most screen time?
- During which periods of the day is smartphone usage highest?

Procrastination

- Is higher screen time associated with higher procrastination levels?
- Which applications contribute most to procrastination-related behaviours?
- How does procrastination vary among students with different screen time levels?

Student Behaviour

- Do students with longer study hours exhibit lower screen time?
- Is there a relationship between sleep duration and smartphone usage?
- Which behavioural patterns are most common among students with high procrastination scores?

Business

- Which student groups would benefit most from digital well-being interventions?
- What features could help students reduce unnecessary screen time and improve academic productivity?

7. Data Collection

Data Source

The dataset used in this case study was collected through a structured Google Forms questionnaire designed specifically for university students aged 20–30 years. The survey gathered information related to smartphone usage habits, academic behaviour, sleep duration, and self-reported procrastination levels.

Sample Size

- Total Responses: **58**
- Target Population: University Students (20–30 years)
- Data Collection Method: Google Forms
- Data Type: Primary Survey Data

8. Data Description

Variable	Type	Description
Age	Categorical	Age group of respondents
Gender	Categorical	Gender
Academic Level	Categorical	UG/PG
Phone Usage	Categorical	Daily screen time
App Category	Categorical	Most used app
Usage Period	Categorical	Morning/Afternoon/Night
Study Hours	Categorical	Daily study duration
Sleep Duration	Categorical	Daily sleeping hours
Procrastination Score	Ordinal	Self-rated score (1–5)

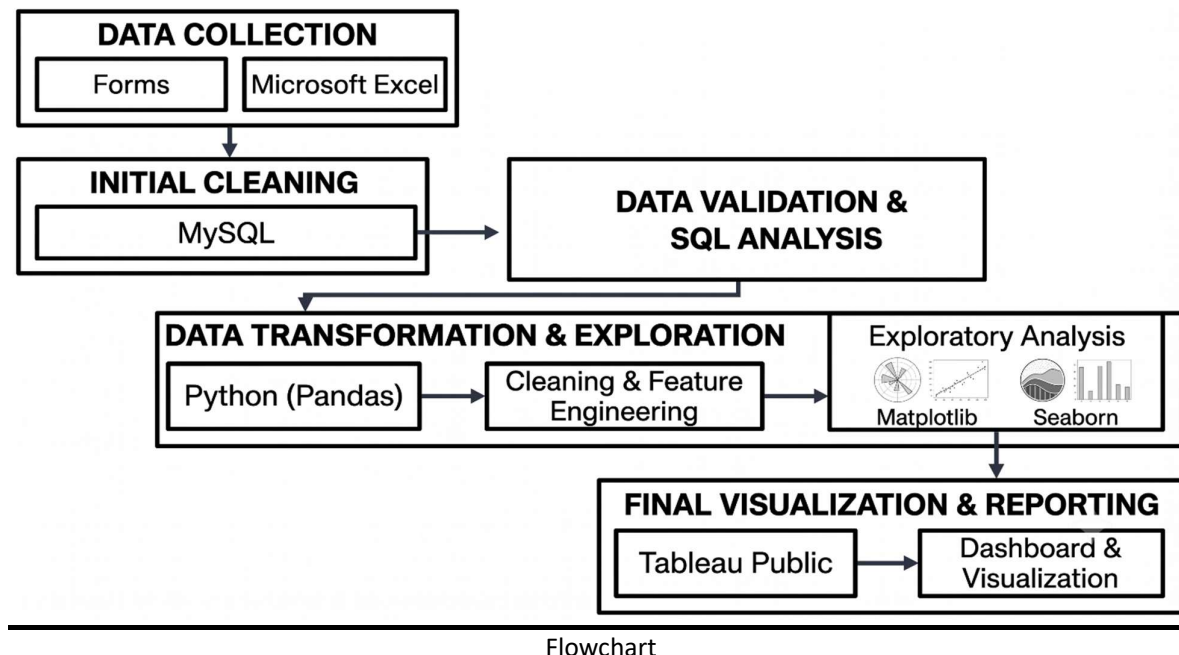
Engineered Variable:

Engineered Variable	Purpose
screen_time_hours	Numeric conversion
study_hours_numeric	Numeric conversion
sleeping_hours	Numeric conversion
phone_check_score	Frequency score
sleep_quality	Derived category
high_distraction	Derived indicator

9.Data Cleaning

- Removed incomplete responses.
- Checked for duplicate records.
- Standardized categorical values.
- Converted categorical ranges into numeric values for statistical analysis.
- Created engineered variables for screen time, study hours, and sleep duration.
- Validated the cleaned dataset using SQL before analysis.

10. Tools Used



11.Data Validation and Exploration Using SQL

The cleaned survey dataset was imported into MySQL Workbench for validation and exploratory querying. SQL was used to verify data integrity, identify missing values, analyse categorical distributions, calculate summary statistics, and prepare the dataset for further analysis in Python and Tableau.

1. Imported dataset into MySQL
2. Verified total number of survey responses

3. Checked for missing values
4. Examined respondent distributions across demographic variables
5. Summarized smartphone usage categories
6. Explored application usage patterns
7. Validated the dataset before Python analysis

12. Exploratory Data Analysis

Data Visualizations in Python

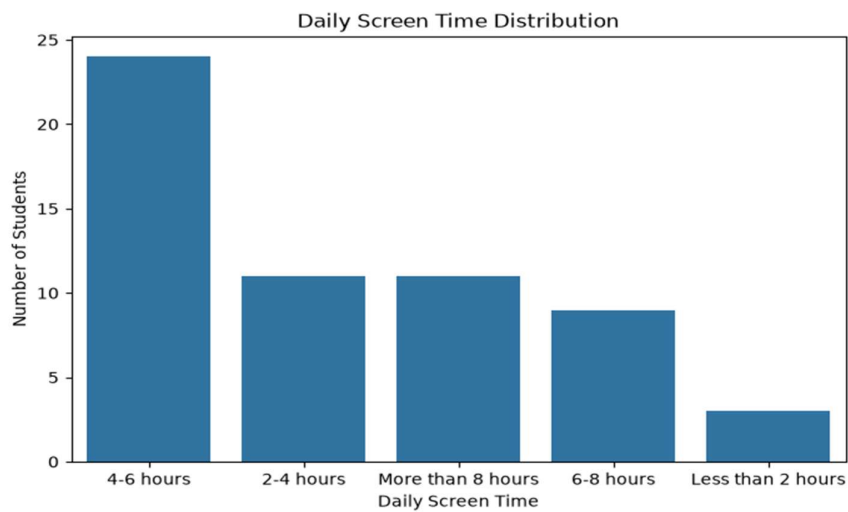


Fig 1

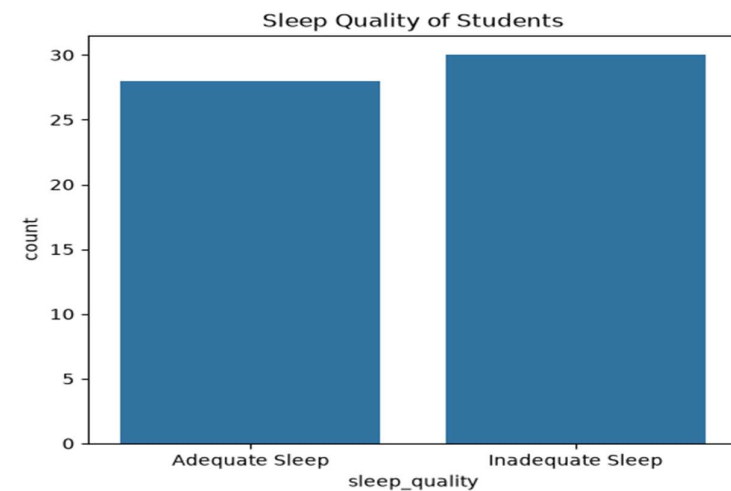


Fig2

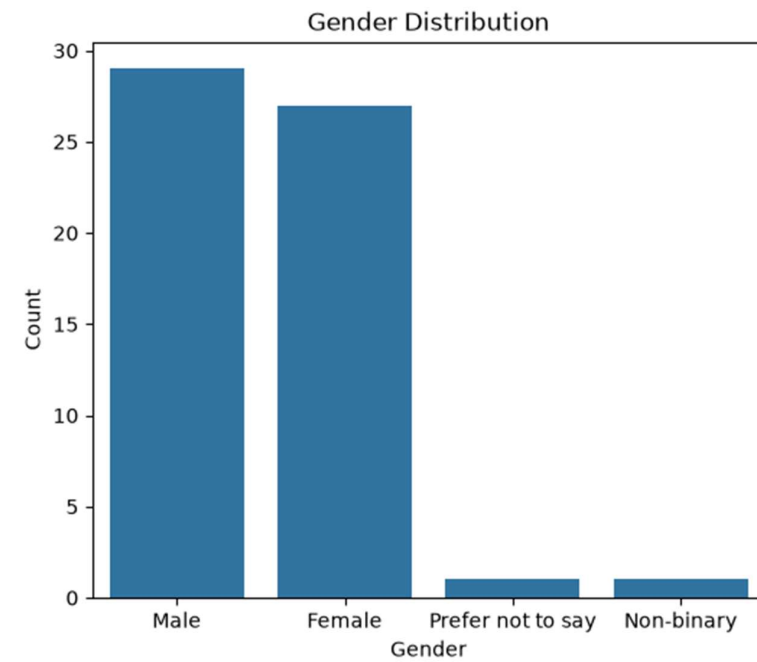


Fig3

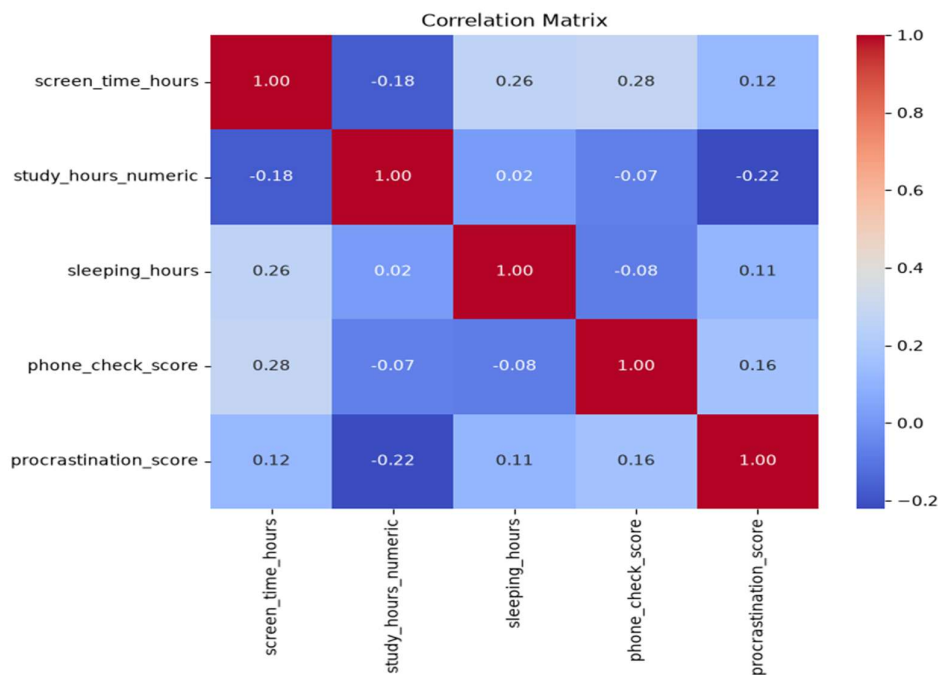


Fig4

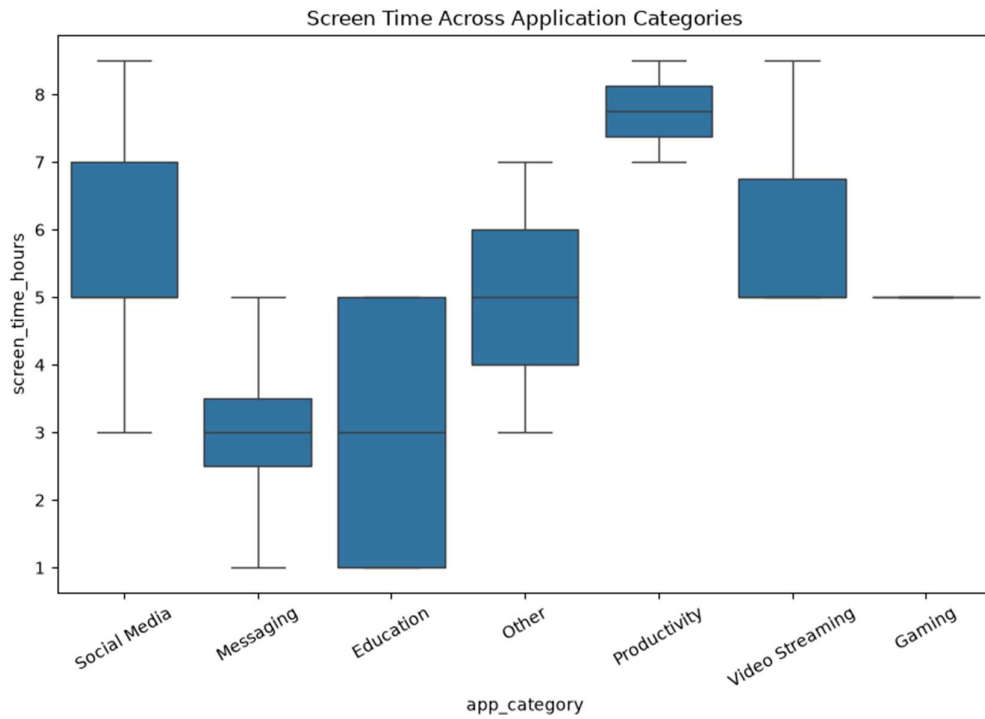


Fig 5

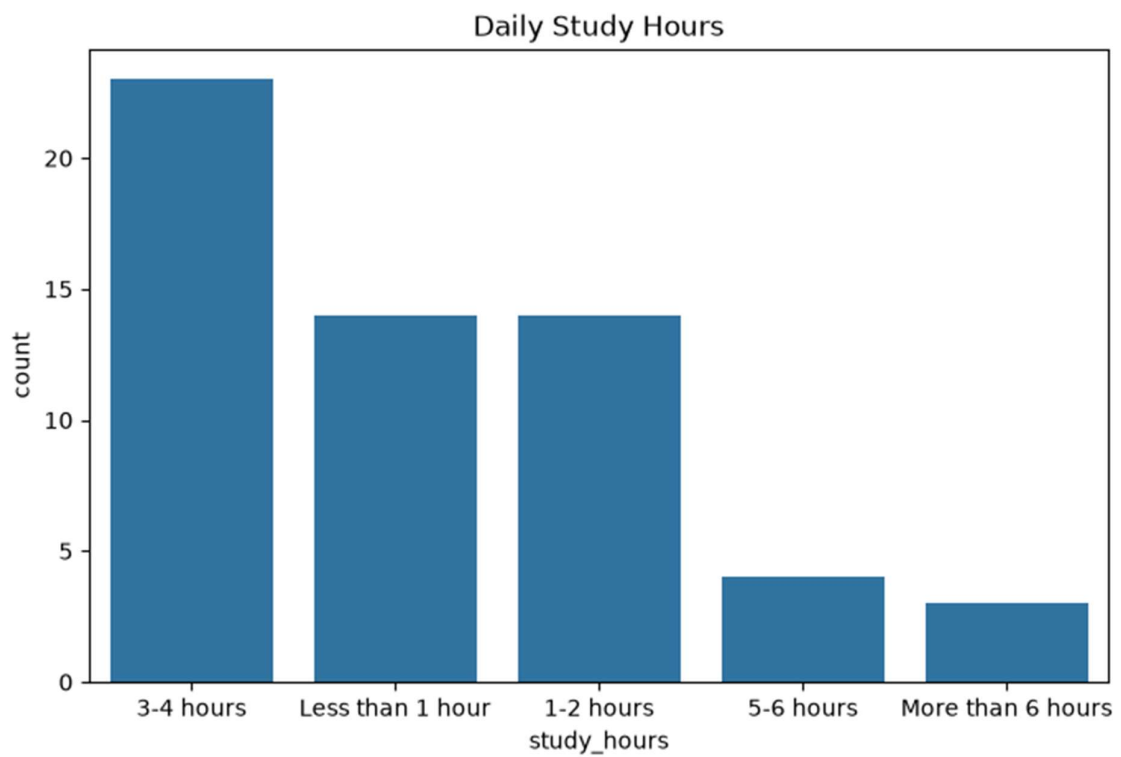
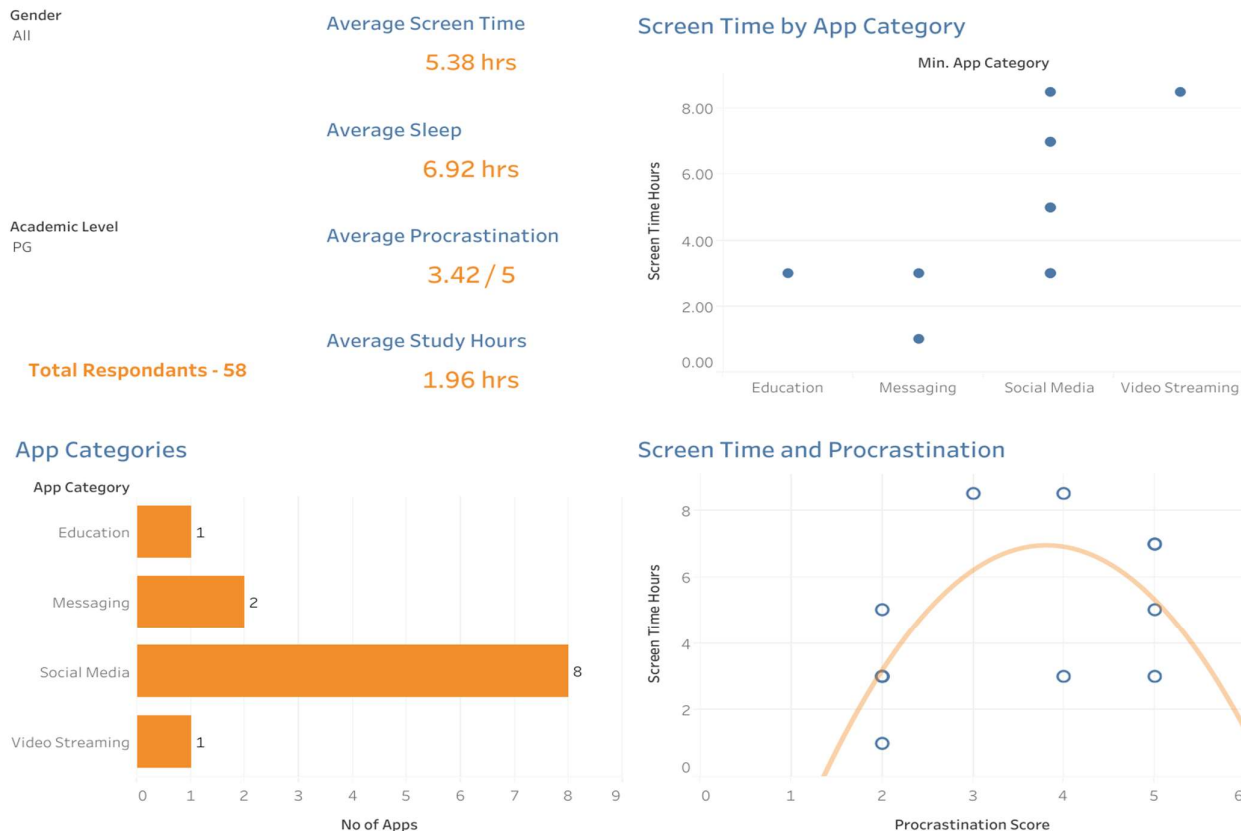


Fig 6

13. Tableau Dashboard

Screen Time & Procrastination Analysis Dashboard



14. Key Findings

- Average daily screen time was 5.39 hours.
- Students studied an average of 2.61 hours/day.
- Average sleep duration was 6.96 hours/day.
- Social media was among the most frequently used application categories.
- Students with higher screen time generally reported higher procrastination scores.
- Average procrastination score was 3.05/5.
- Most respondents belonged to undergraduate programs.

15. Business Recommendations

Based on the findings of this case study, the following recommendations are proposed for FocusTech to enhance its digital well-being platform and improve student productivity.

1. Implement Personalized Screen Time Dashboards

Develop personalized dashboards that monitor daily smartphone usage and provide automated alerts when users exceed recommended screen time limits. Personalized usage insights can encourage students to become more aware of their digital habits and promote healthier screen time management.

Expected Business Value

Encourages self-awareness and healthier digital habits.

Increases user engagement through personalized feedback.

Strengthens user retention by providing actionable insights.

2. Introduce an AI-Powered Smart Focus Mode

Develop an intelligent Focus Mode that temporarily blocks distracting applications and websites during study sessions while integrating productivity techniques such as the Pomodoro method. This feature can help students maintain concentration and reduce interruptions.

Expected Business Value

Improves study efficiency and academic productivity.

Creates opportunities for premium subscription features.

Differentiates FocusTech from competing productivity applications.

3. Deliver Weekly Productivity Insight Reports

Generate personalized weekly reports summarizing key productivity indicators such as screen time, study duration, sleep patterns, and overall productivity trends. Providing regular feedback can motivate users to monitor their progress and adopt healthier digital behaviours.

Expected Business Value

Promotes long-term user engagement.

Encourages consistent application usage.

Supports data-driven self-improvement.

4. Expand Sleep-Aware Digital Wellness Features

Introduce sleep-aware functionality by monitoring late-night smartphone usage, providing bedtime reminders, and automatically reducing unnecessary notifications during sleeping hours. This would encourage healthier sleep routines while supporting overall digital well-being.

Expected Business Value

Improves user well-being beyond screen time management.

Positions FocusTech as a comprehensive digital wellness platform.

Enhances customer satisfaction through holistic lifestyle support.

5. Develop an AI-Based Procrastination Detection System

Utilize machine learning models to identify behavioural patterns associated with procrastination, including excessive screen time, frequent phone checking, and prolonged social media usage. The system can proactively recommend personalized focus sessions and productivity interventions.

Expected Business Value

Enables proactive rather than reactive user support.

Enhances personalization through predictive analytics.

Strengthens the platform's AI-driven capabilities.

6. Introduce Gamification to Encourage Healthy Digital Habits

Incorporate gamification features such as productivity challenges, achievement badges, study streaks, and milestone rewards to motivate students toward healthier smartphone usage patterns.

Expected Business Value

Increases daily active users (DAU).

Improves long-term user retention.

Encourages positive behavioural change through engagement.

7. Implement User Behaviour Segmentation

Segment users into behavioural profiles such as Productive Learners, Balanced Users, and High-Risk Procrastinators. Personalized recommendations and interventions can then be delivered based on each user's behavioural profile.

Expected Business Value

Supports highly personalized user experiences.

Improves recommendation accuracy.

Enables targeted feature development and marketing strategies.

8. Expand Predictive Analytics Capabilities

Future versions of the application should incorporate additional behavioural data, including real-time screen activity, notification frequency, assignment schedules, and study deadlines. These additional data sources can improve predictive modelling and support more intelligent recommendations.

Expected Business Value

Improves prediction accuracy.

Supports continuous product innovation.

Enables more advanced AI-powered digital well-being solutions.

Overall Business Impact

Implementing these recommendations would enable FocusTech to evolve from a screen time monitoring application into an intelligent digital well-being platform. By combining behavioural analytics, personalized recommendations, and AI-driven productivity features, the company can help students reduce digital distractions, improve study habits, and enhance academic performance while increasing user engagement, retention, and long-term product value.

16. Limitations

- The study was conducted using a relatively small sample of 58 university students, which may not fully represent the broader student population.
- The dataset was based on self-reported survey responses, making it subject to recall bias and individual interpretation.
- The analysis focused primarily on students aged 20–30 years, limiting the generalizability of the findings to other age groups.
- As this was a cross-sectional study, the analysis identifies associations rather than establishing causal relationships between screen time and procrastination.

17. Future Scope

- Collecting responses from a larger and more diverse student population.
- Integrating real-time smartphone usage data instead of self-reported information.
- Developing machine learning models to predict procrastination risk.
- Expanding the dashboard with real-time analytics and personalized recommendations.
- Conducting longitudinal studies to examine changes in digital behaviour over time.

18. Conclusion

This case study explored the relationship between smartphone screen time and procrastination among university students using an end-to-end data analytics workflow. Data collected through a structured survey were validated using SQL, analysed in Python through exploratory data analysis and feature engineering, and visualized using an interactive Tableau dashboard.

The findings indicate that extended smartphone usage is associated with increased procrastination and highlight opportunities for promoting healthier digital habits. Based on these insights, several business recommendations were proposed for the hypothetical EdTech company FocusTech, including personalized screen-time monitoring, intelligent focus tools, and AI-driven productivity features.

Overall, this project demonstrates how data analytics can transform survey data into meaningful insights that support evidence-based decision-making and digital well-being initiatives.